

CS795 Course Project Proposals

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1) TinyLLM: Evaluating Post-Training Quantization on a Small LLM

Background: Large Language Models (LLMs) are powerful but computationally expensive, making deployment on laptops or edge devices challenging. This project focuses on post-training quantization (PTQ), a popular approach for reducing model size and improving inference speed without retraining.

Goals: Apply, evaluate, and compare two leading PTQ methods (GPTQ, AWQ) on the same open-source model (e.g., Phi-3-mini or Mistral-7B), creating fast, CPU-runnable versions while quantifying accuracy and efficiency trade-offs.

Data / Benchmarks: WikiText-103 for perplexity; small subsets of MMLU, ARC-Easy, GSM8K; calibration set of ~1024 samples from C4/WikiText.

Technical Components: Model + dataset downloader, quantization pipeline (GPTQ & AWQ), evaluation harness (memory, latency, throughput, perplexity, accuracy), packaging models to GGUF for llama.cpp demo.

Expected Output: Comparison table of model size, perplexity, accuracy, latency, memory footprint; notebook + CLI demo for local inference.

Stretch Goal: Investigate SparseGPT pruning as a complementary compression step.

2) Edge-RAG: A Private, On-Device QA System for Documents

Background: Organizations in sensitive fields such as healthcare often need answers from private documents but cannot use cloud LLMs. This project builds a fully self-contained, offline RAG pipeline running entirely on a laptop.

Goals: Demonstrate an end-to-end private QA system with local retrieval, quantized reader model, and measurable accuracy, all offline.

Data / Benchmarks: Public PDFs (CMS fact sheets, benefit docs); 50–100 labeled QA pairs; metrics: recall@k, EM/F1, latency (<2s).

Technical Components: PDF ingestion + chunking to FAISS, retriever (MiniLM), quantized reader model (Phi-3-mini-4k-instruct-q4.gguf), evaluation harness (EM/F1), CLI demo.

Stretch Goal: Experiment with re-ranking or chunk-size tuning for improved recall.

References

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